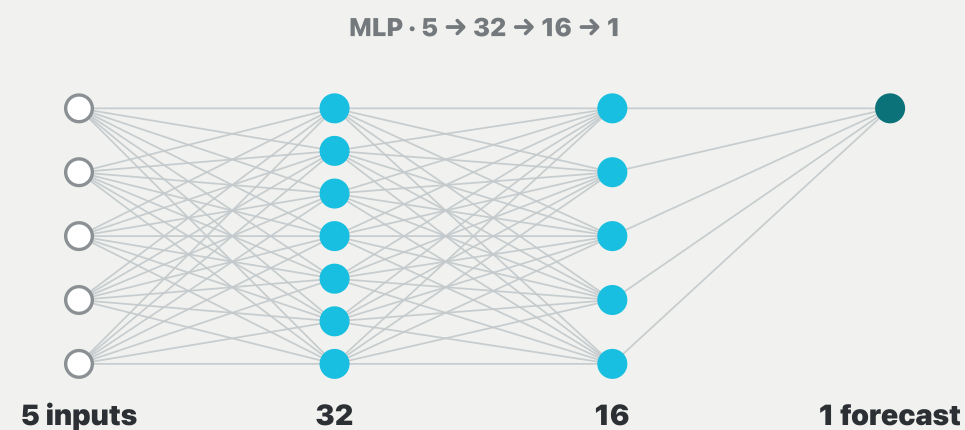


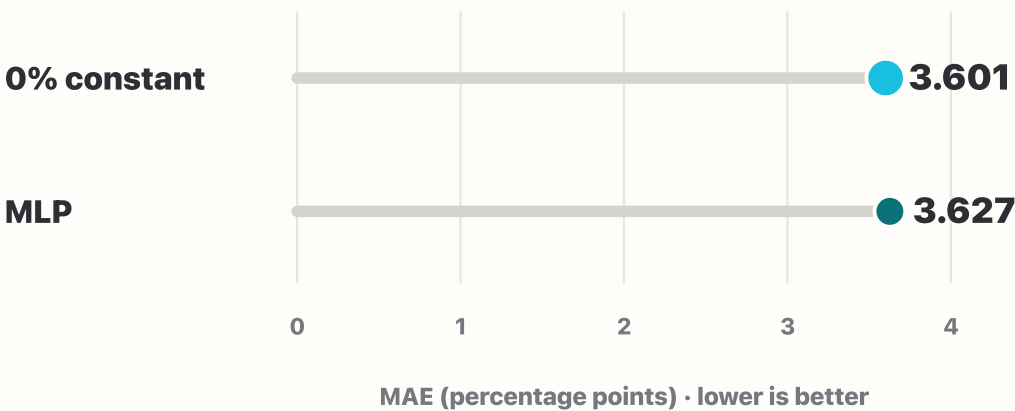
# I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.  
"Predict 0% growth" still beat it on mean absolute error (MAE).



TARGET: NEXT-QUARTER DOMESTIC DEPOSIT GROWTH

|                |                     |
|----------------|---------------------|
| Deposits       | \$100m → \$96m      |
| Actual growth  | -4%                 |
| Zero forecast  | 0%                  |
| Absolute error | 4 percentage points |



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

# YOUR DEPOSITS HELP FUND LOANS. THE BANK EARNs INTEREST.

THE BANK DOESN'T KEEP EVERY DEPOSIT DOLLAR IN CASH.



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

A customer withdraws \$15. Cash is \$5 short.

THE BANK'S TREASURY TEAM MUST FIND THE CASH.

HOW MUCH CASH IS ENOUGH?



Required reserve ratio: 0%. That does not mean unlimited lending.

There is no universal “keep 10%, lend 90%” rule. Capital, liquidity and other limits constrain lending.

The full liquidity coverage rule requires eligible liquid assets covering 100% of 30-day stressed net cash outflows.

CASH

DEPOSITS

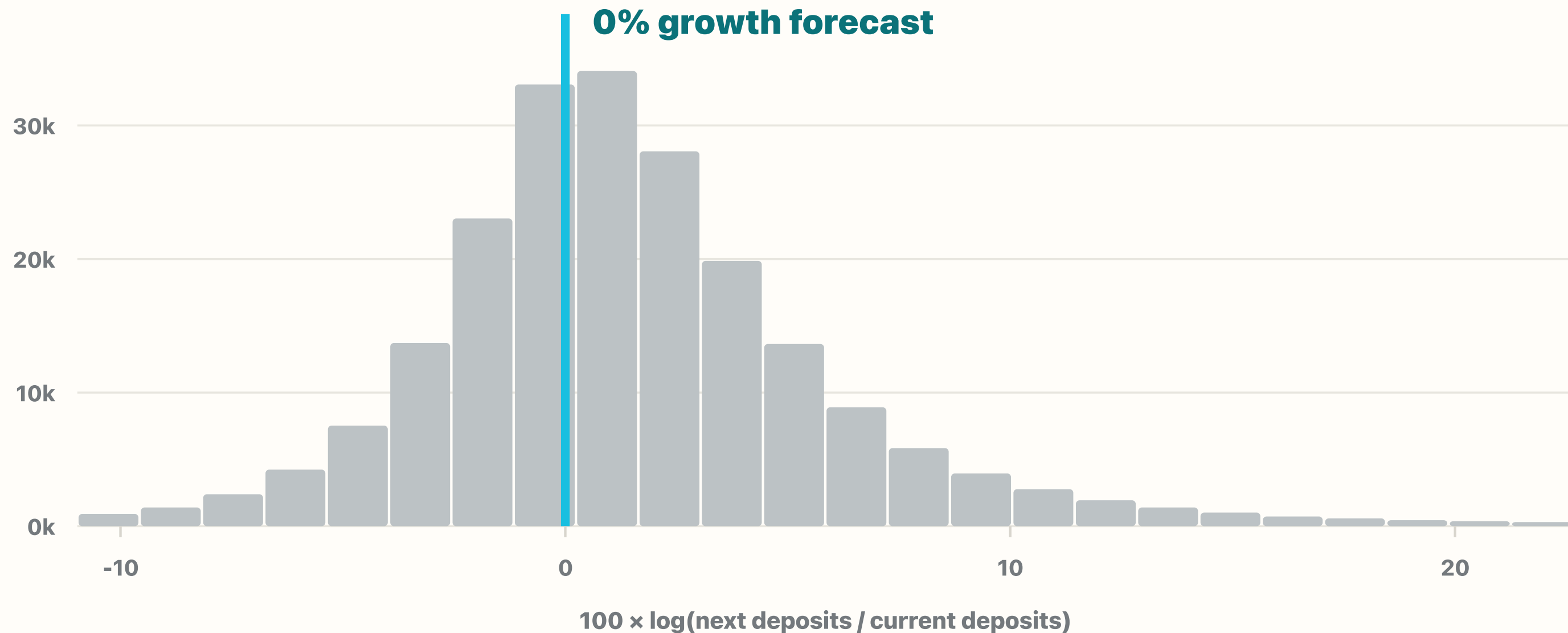
LOANS

CAPITAL

CUSTOMERS DEPOSIT AND WITHDRAW MONEY.  
I'M FORECASTING THE NET CHANGE NEXT QUARTER.

Example only. Reserve requirement ratios are 0%; separate liquidity rules vary by bank. Liquid assets include eligible securities, not just cash. Source: Federal Reserve, Regulation WW. This project forecasts deposit growth, not bank failure.

# SO WHY WAS 0% GROWTH SO HARD TO BEAT?



**A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.**

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

# BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

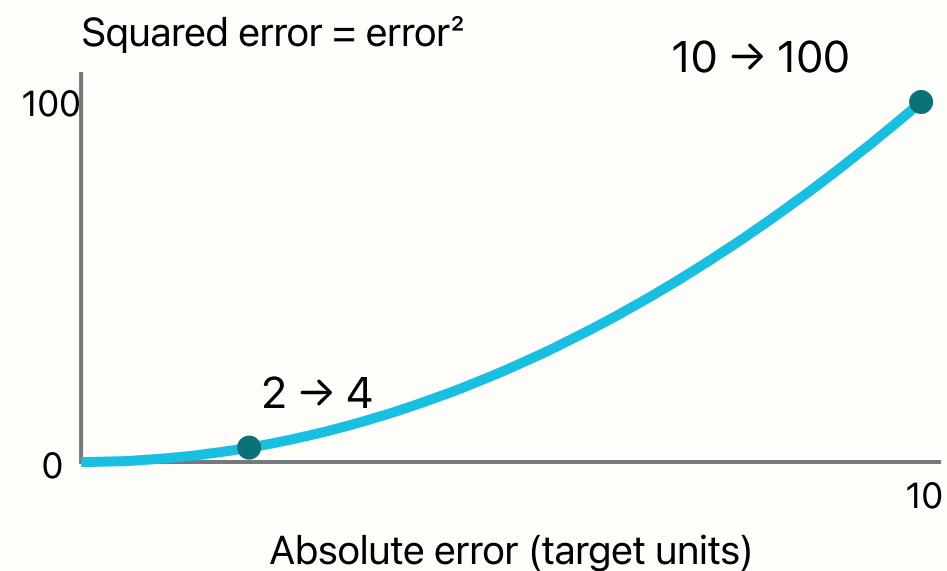
An error 5× as large gets 25× the squared-error penalty.

## FIXED BENCHMARK

### 0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.  
It never adjusts to reduce an error.



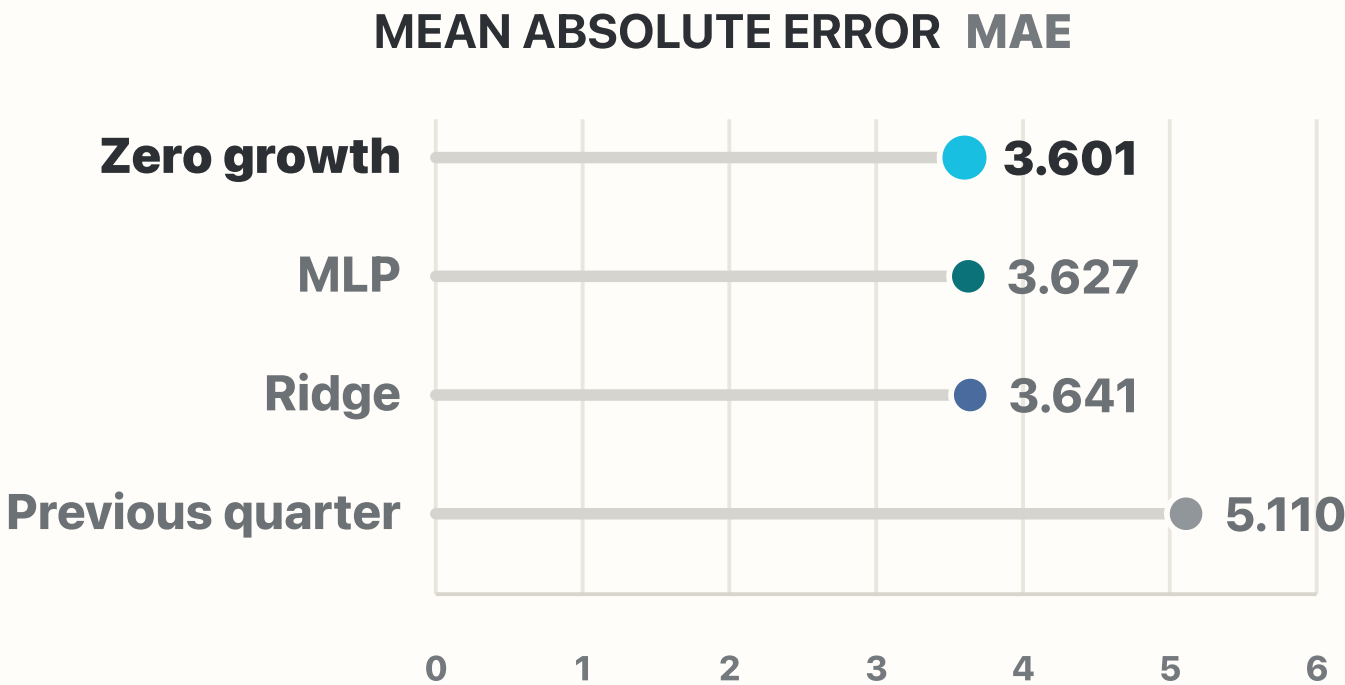
## LEARNED MODELS

### RIDGE + NEURAL NET

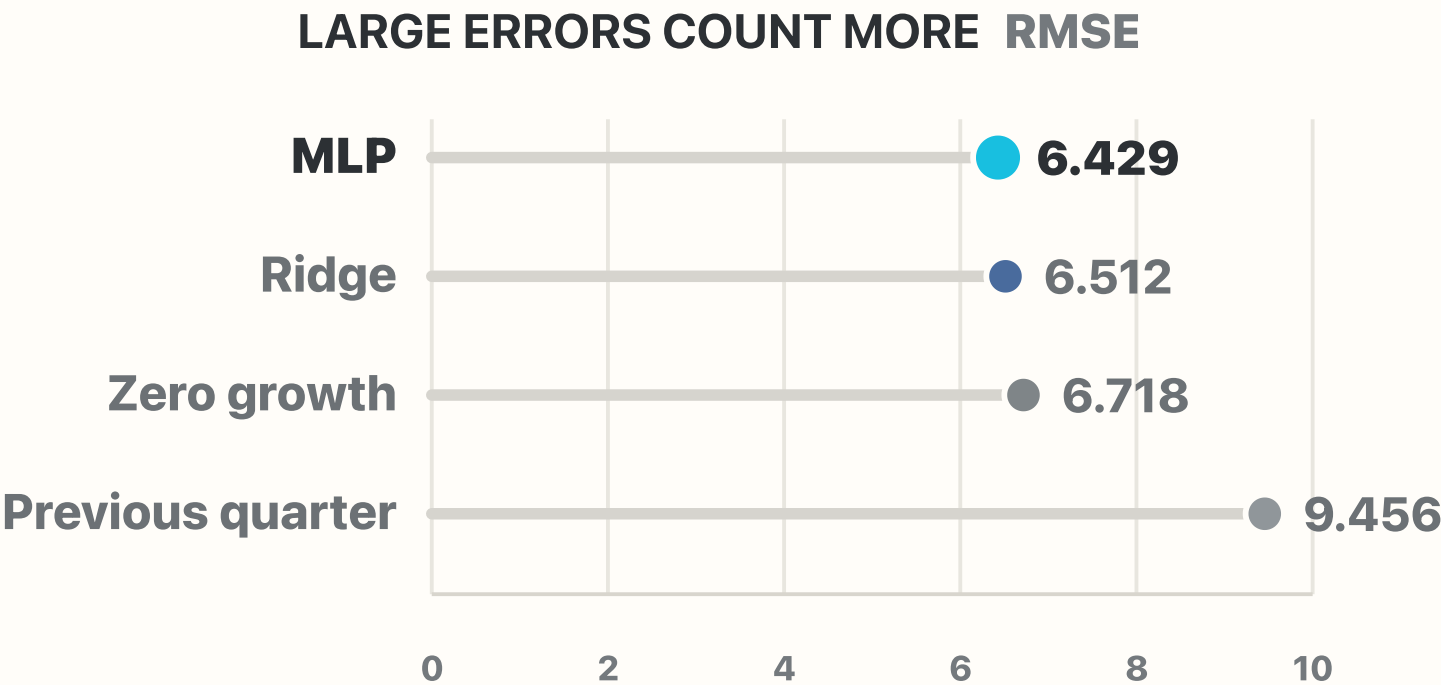
Can move away from 0% to reduce  
those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH  
A SLIGHTLY HIGHER MEAN ABSOLUTE ERROR.**

# ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



MAE (percentage points) · lower is better



RMSE (percentage points) · lower is better

**“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.**  
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

# 0% TIES EVERY BANK.

## THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

★ I am not a financial advisor. I do have a sorted table.

In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%. Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

| RANK | BANK                       | PREDICTED GROWTH | 0% BASELINE |
|------|----------------------------|------------------|-------------|
| 01   | YOUNG AMERICANS BANK       | -4.02%           | 0% · tied   |
| 02   | PIONEER COMMERCIAL BANK    | -3.32%           | 0% · tied   |
| 03   | SUMITOMO MITSUI TR BK U SA | -2.83%           | 0% · tied   |
| 04   | OLD GLORY BANK             | -2.75%           | 0% · tied   |
| 05   | TURTLE MOUNTAIN STATE BANK | -2.11%           | 0% · tied   |
| 06   | COLUMBUS STATE BANK        | -2.05%           | 0% · tied   |
| 07   | FDS BANK                   | -2.03%           | 0% · tied   |
| 08   | BOELUS STATE BANK          | -1.96%           | 0% · tied   |
| 09   | LEMONT NATIONAL BANK       | -1.80%           | 0% · tied   |
| 10   | CITIZENS NB OF CROSBYTON   | -1.75%           | 0% · tied   |

GOAL: FIND MORE LOWEST-GROWTH BANKS WITH THE SAME 454 REVIEWS.

HIGHER IS BETTER.

Random selection

45.4 expected

Linear model (Ridge)

112 found

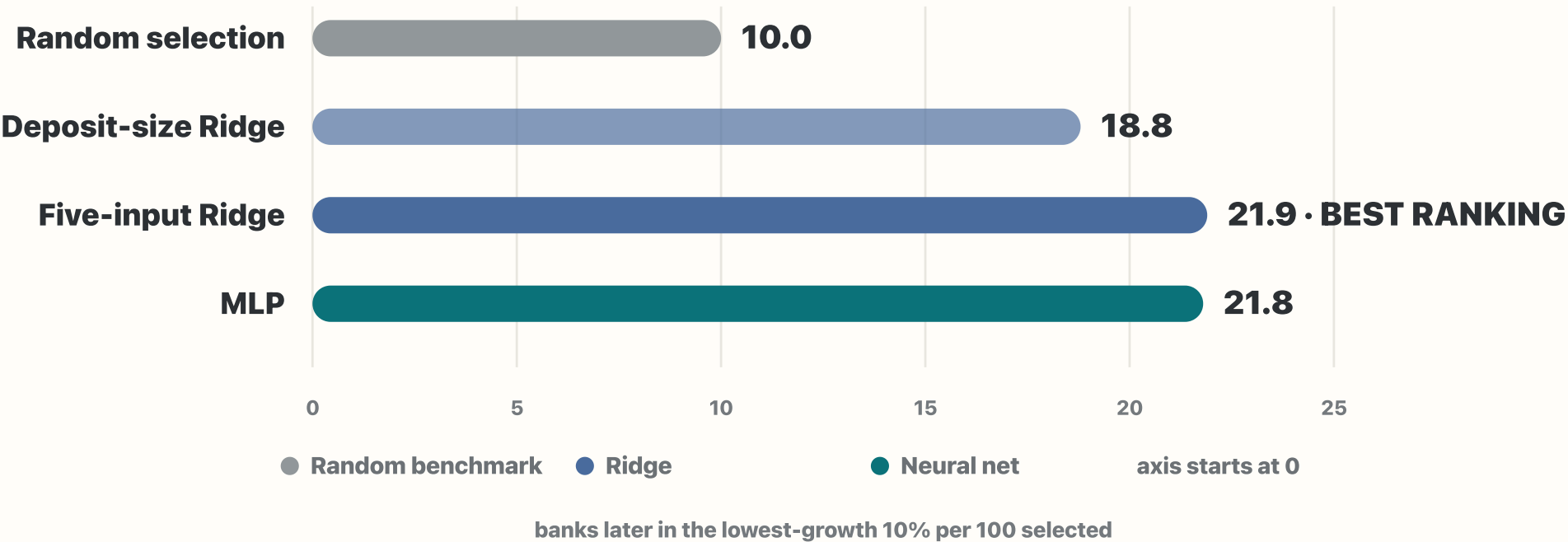
Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth. 112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

# I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

I tested deposit size alone to see whether four extra inputs actually improved the review list.



**MAE WINNER: PREDICT 0% → 3.601.** Deposit-size Ridge: 3.654 · Five-input Ridge: 3.641 · Neural net: 3.627.

All three lost to zero on MAE. Zero ties every bank, so it cannot order a review list.

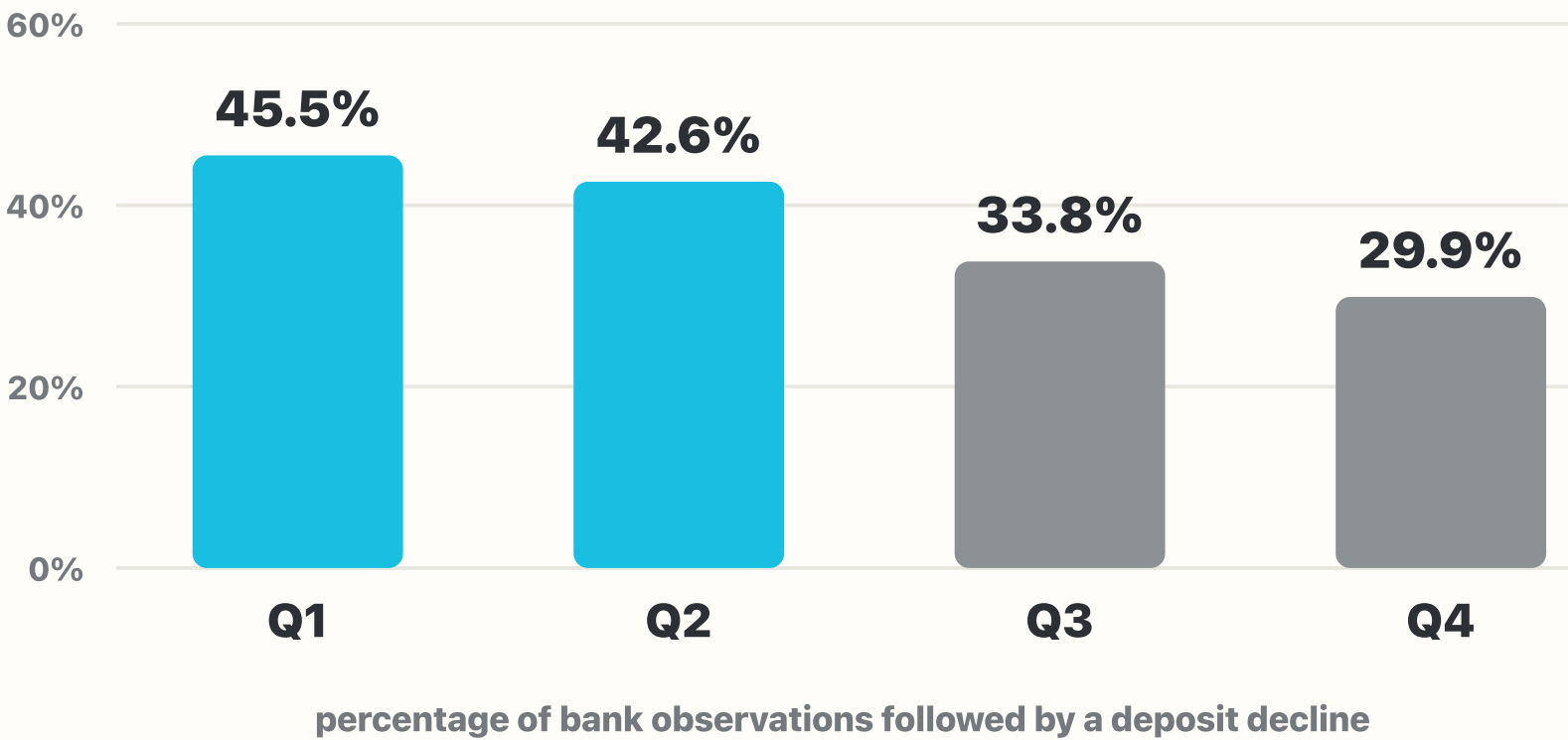
**DEPOSIT SIZE ALONE: 18.8 FOUND PER 100 REVIEWED.**  
**ALL FIVE INPUTS: 21.9. A GAP OF 3.1.**

# DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

**Q1:** 45.5% of training bank-quarters were followed by a deposit decline. **Q4:** 29.9%.

That is how often declines happened, not “banks lost 45.5% of their deposits.”

HOW OFTEN DID A BANK’S DEPOSITS FALL NEXT QUARTER? · TRAINING YEARS 2013–2022

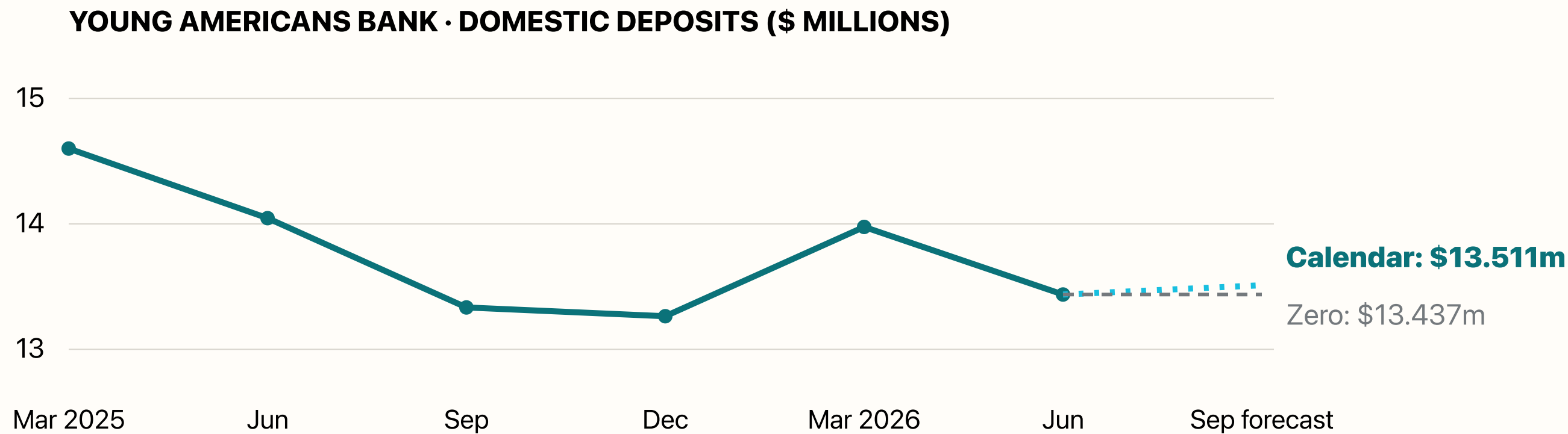


THE STARTING QUARTER MAY HELP SET A GROWTH FORECAST.

For each bar, count bank observations with a next-quarter decline, then divide by all observations starting in that quarter (training data through Sep 2022). Deposit dollars are not added together.

# MODEL CHOSEN FOR THE NEXT TEST: THE CALENDAR RULE. NOW WE WAIT FOR Q3 REPORTS.

Q3 ends September 30, 2026. Reports arrive afterward. The saved forecast is +0.550% per bank; zero is the benchmark.



Solid = reported balances. Dotted = calendar forecast (+0.550%). Dashed = no-change forecast.

**WHEN SEPTEMBER REPORTS ARRIVE, COMPARE MEAN ABSOLUTE ERROR.  
SAME BANKS. SAME QUARTER. LOWER MAE WINS.**

FDIC BankFind, CERT 27010, retrieved Sep 22, 2026. One-bank illustration of the saved Q2 median rule (+0.550399842%); test MAE across matched banks. Rule developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.